Unveiling Prognostic Pattern: A Hybrid Approach For Predictive Analytics in Medical Image Data

Uttkarsh Parmar
Computer Science and Enginnering Department, M.tech
Abes enginnering college
Ghaziabad, India
uttkarshparmar50@gmail.com

Anil Kumar Dubey
Faculty of Computer Science and Enginnering Department
Abes enginnering college
Ghaziabad, India
anil.dubey@abes.ac.in

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***Abstract*— Currently, medical image data are being processed using deep learning technology to revolutionize disease-related research. Deep learning-based medical image analysis involves the application of various essential tasks, including medical image reformation, enhancement of image quality, segmentation, classification, and Computer-Aided Detection (CAD). These tasks aim to discern differences between healthy cells and tumour cells, facilitating more accurate diagnoses and effective treatment planning. This paper focuses on employing a hybrid approach for medical image analysis, integrating both clustering and classification techniques. The proposed methodology leverages k-means clustering, Visual Geometry Group (VGG), and Recurrent Neural Network (RNN) on distinct datasets related to cardiovascular diseases, breast cancer, and brain diseases. The hybrid approach aims to capitalize on the strengths of both clustering and classification methods, enhancing the overall accuracy and interpretability of the medical image analysis process. By utilizing diverse datasets, the research endeavours to contribute valuable insights into the identification and understanding of various diseases, ultimately advancing the field of medical image analysis for improved healthcare outcomes.**

Keywords*— *Medical image analysis. Predictive Analytics, Medical Image Data, Classification, Clustering, VGG, RNN, Hybrid Approach, Feature Extraction, Machine Learning, Healthcare.

I. INTRODUCTION

Medical imaging, the process of capturing internal organ images for diagnostic and research purposes, plays a crucial role in disease detection and study, aiming to enhance the effectiveness of clinical research and treatment options. The demand for medical imaging services has witnessed a significant upsurge, particularly fuelled by the impact of the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) discovered in late 2019. Various types of medical images, including X-rays, endoscopies, CT scans, mammography images, ultrasounds, MRI, MRA, nuclear medical imaging, PET scans, and pathology tests, contribute to a comprehensive diagnostic approach.

However, the interpretation of medical images can be challenging and time-consuming, given the shortage of radiologists. The advent of Artificial Intelligence (AI) emerges as a promising solution to address these challenges. AI technologies, particularly machine learning and deep learning algorithms, can be employed to automate and expedite the analysis of medical images. This not only aids in overcoming the limitations posed by the scarcity of radiologists but also enhances the accuracy and efficiency of diagnostics. Machine learning (ML) stands as a formidable application of artificial intelligence (AI) capable of autonomous functioning without explicit programming, relying on acquired knowledge from data to make predictions or decisions based on historical information [3].

However, the advent of Deep Learning (DL), a subset of ML, has revolutionized this paradigm by addressing the challenge of feature selection. The defining characteristics of DL include the incorporation of multiple processing layers, allowing the algorithm to discern discrete features at varying levels of abstraction. Moreover, DL facilitates both unsupervised and supervised learning of feature representations at each layer, providing a powerful and flexible framework for addressing complex problems in diverse domains.

A. History of medical images.

Medical imaging was first commenced with the discovery of X-rays in 1895 by Wilhelm Conrad Roentgen's. After that on 11 November 1896 first Radiology was introduced by John Hall Edward (Who was the well renowned surgeon and licensed to practice medicine by the Royal College of Physicians of Edinburgh, in Scotland), When he first radiographed his associate hand and identifies the sterilized needle beneath the exterior part of his skin. Then after on 14 February he used radiography in surgical operation and took first human spin X-ray.

In 1927, Portuguese physician Egas Moniz pioneered angiography, revolutionizing the observation of nervous diseases through contrasted cerebral imaging. Angiography involves injecting a contrast agent to enhance X-ray visibility of blood vessels, heart chambers, or organ lumens, aiding in the identification of deformities. This process, known as fluoroscopy, enables precise visualization of internal structures.

After the Angiography, In 1953 Ultrasonography is introduced in this world by scientists named I. Elder and C. Hertz. Ultrasonography does not use any X-ray, it uses the high frequency wave which was passes through the device called Transducer. Transducer convert those electric waves (current) into high frequency sound wave (ultrasound) and transmitted into the tissues of the human body, after travelling through tissues the sound wave bounce back to the

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transducer. which converted those sound waves into the electric signal and those patterned electric signal converted into the digital image by the computer.

used resistive magnetic fields for low spatial resolution images, but modern MRI employs hydrogen nuclei due to their abundance in water and unique spin properties.

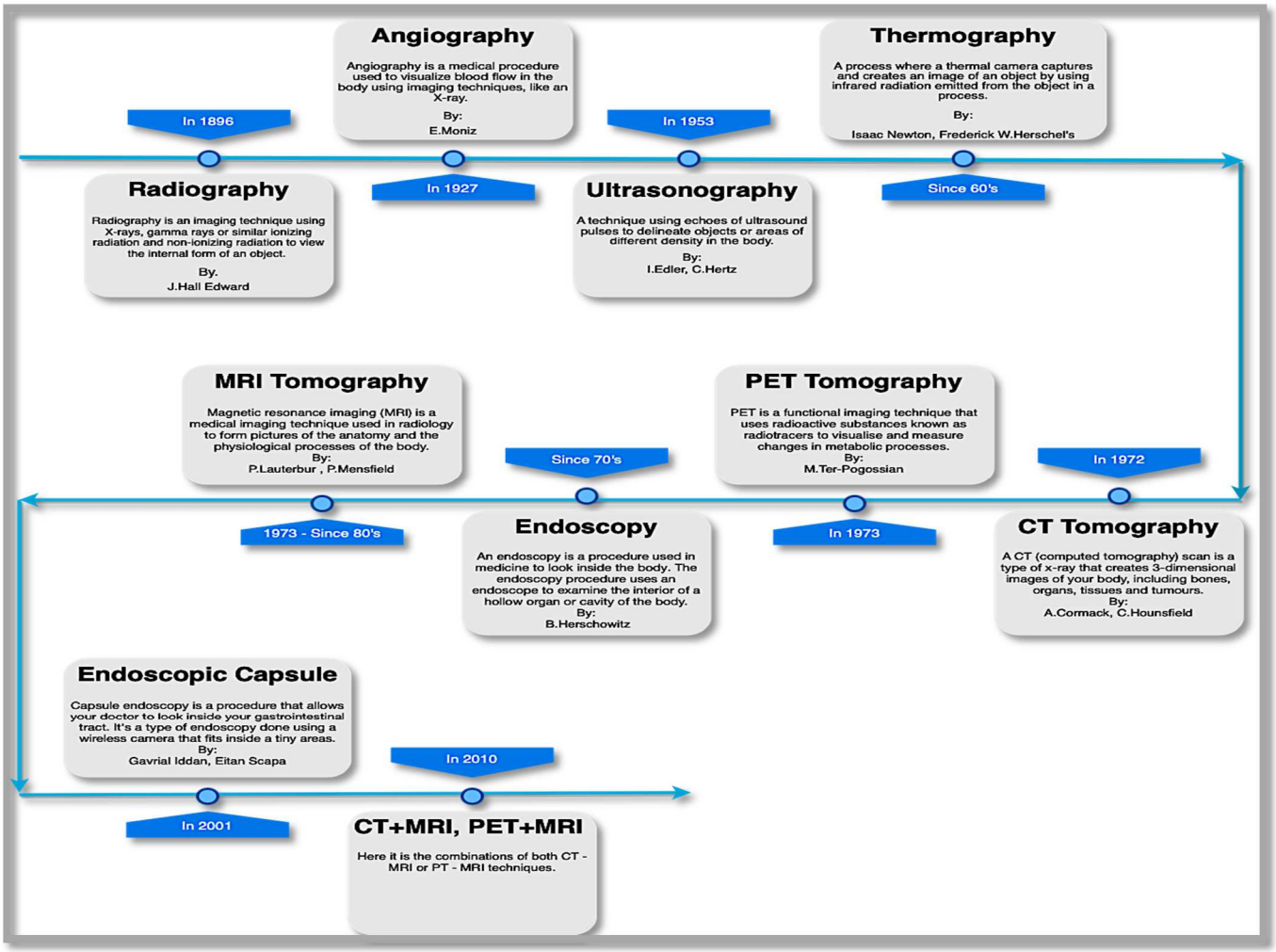


Fig. 1. Above diagram shows the history of medical image.

In 1972, physicist Allen Macleod Cormack and engineer Godfrey Hounsfield collaborated to introduce CT Tomography. This technique utilizes X-rays, emitted towards an object and then captured as they bounce back to create images. Notably, CT Tomography distinguishes itself by producing 3D images through the rotation of the X-ray tube and detectors in a gantry. Widely employed in medical diagnostics, it detects brain tumors, thyroid cancer, coronary diseases, and assesses changes in lung parenchyma, offering a versatile imaging solution.

Physicist M.Ter–Pogossian introduced PET Tomography, employing positron emission for imaging in nuclear medicine. In PET, positron-emitting isotopes like F-18 are used with fluorodeoxyglucose (FDG) to visualize increased glucose uptake in cancer. PET-CT combines PET's metabolic information with CT's high resolution, facilitating precise cancer localization for biopsy, radiation therapy, or surgery. This technique has become crucial in diagnosing both primary and metastasized cancers, offering an advanced and comprehensive imaging approach.

Following endoscopy, MRI Tomography emerged in the 1970s for non-invasive imaging of human organs. Initially, it

The resonance frequency of hydrogen nuclei is magnetic field-dependent, emitting radiofrequency signals when resonating. Detection by an RF coil around the affected body part produces detailed images, facilitating diagnosis and treatment planning.

Over the past 120 years, medical imaging has evolved significantly since the advent of X-rays. The integration of hybrid methods, such as PET+MRI and CT+MRI, has further advanced tomography for diagnosing medical issues related to patients' organs. These hybrid approaches provide more accurate and advanced information, facilitating early diagnosis and guiding necessary exploratory surgery.

B. Motivation

Medical imaging is pivotal in modern healthcare, aiding clinicians in precise diagnosis and treatment planning by visualizing internal structures. This research paper delves into recent advances in imaging analysis, emphasizing novel techniques for enhanced disease detection, classification, and prognosis. The critical significance of accurate and timely diagnosis is highlighted for improved patient outcomes and healthcare efficiency.

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Nowadays, the researcher's work on the Medical image analysis is only focus on the particular topic (like- They only using the classification for the detection of the disease on the pre-processed data and labeled data), through which they make the identification of the disfunction of the patient's organ through the medical images and also didn't care about 'how to use that un-labeled or labeled data in the dataset to the proposed model'. But we will uses both non pre-processed and non labeled data for our model, which can reduces the headache of the examiner of feeding the only labeled data to the model.

A hybrid approach in medical image analysis is motivated by the synergy between classification and clustering methods. Classification excels in labeling predefined categories like disease/no disease, while clustering unveils inherent patterns in data structures. Combining these methodologies in a hybrid model allows for capturing both categorical information and underlying structures in medical images, enhancing predictive accuracy and robustness.

Complexity of Medical Image Data: Medical images are inherently complex, often containing intricate patterns, variations, and subtleties that are challenging to capture using traditional analytical methods. The motivation behind the hybrid approach lies in recognizing the diverse and multifaceted nature of medical image data, aiming to leverage both classification and clustering to unravel the complexities embedded within the images. That's why we use two types of dataset from Kaggle (labeled data) and IEEE data port (unlabeled data).

Complementary Nature of Classification and Clustering: Classification models (Through VGG, RNN, Decision Tree, Random Forest) excel in assigning predefined labels to distinct classes, while clustering algorithms (Through reveal hidden structures within data. The motivation stems from the acknowledgment that these methodologies are complementary. By combining their strengths, the hybrid model seeks to harness the discriminative power of classification alongside the unsupervised learning capabilities of clustering, offering a more holistic understanding of the underlying information.

Improved Predictive Accuracy and Precision: The primary motivation is to enhance predictive accuracy in medical image analysis. The hybrid model aspires to provide more precise predictions by simultaneously considering explicit class labels and latent structures within the data. This approach is driven by the goal of producing more nuanced and accurate results, thereby improving the reliability of diagnostic and prognostic outcomes.

Handling Heterogeneity Through Clustering in Medical Datasets: Medical datasets often exhibit significant heterogeneity due to variations in patient populations, imaging technologies, and disease manifestations. The motivation for the hybrid approach lies in its potential to effectively handle such heterogeneity, accommodating diverse data characteristics and enhancing the model's adaptability to the complex and varied nature of real-world medical imaging data. Because in real world data some data are distorted, which is neglected or not having used in the dataset but it have some information hidden in it. So we use clustering (Through K-mean) to use that distorted data with same property having data and making cluster with data.

Early Detection and Personalized Treatment Planning: Timely disease detection and personalized treatment are crucial for better patient outcomes. The hybrid model aims to identify subtle patterns indicative of early-stage diseases, facilitating more effective interventions and aligning with the goal of enhancing patient care through tailored medical strategies.

C. Why we use clustering in our model?

Here we use clustering only for the labeling of the data because in medical data 10 percent data out of the 100 percent are unlabeled or we can say that, that data is unknown and does not belong to any particular column (attribute). Here we apply Pseudo- labeling for the labelling of the image on the basis of similarity.

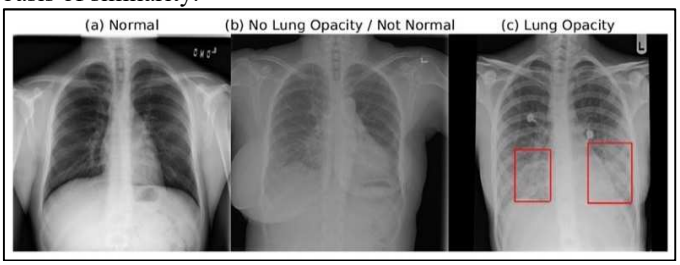


Fig. 2. a) Shows normal chest X-ray. (b) Shows no particular class of diseases or we can say that, it is not normal chest X-ray. (c) Shows the symptoms of pneumonia.

The motivation for adopting a hybrid approach in predictive analytics for medical image data is rooted in addressing the complexities inherent in healthcare datasets, leveraging the synergies between classification and clustering, and ultimately striving to provide more accurate, interpretable, and clinically valuable insights for improved patient care.

II. Literature review

Over the last decade, predictive analytics in medical image data has advanced, focusing on methodologies to improve diagnostic accuracy. This review chronicles key contributions from 2010 to 2022, emphasizing hybrid models' development, key features, and challenges encountered in medical image analysis.

A. Review

The literature survey on medical image analysis provides a comprehensive and insightful overview of the field's evolution, capturing key themes, methodologies, and challenges. The review adeptly navigates through different eras of research, starting with traditional image processing techniques and progressing to the current era dominated by machine learning and deep learning approaches.

Notably, the inclusion of foundational works serves to contextualize the historical progression of research [1]. The emergence of computer-aided diagnosis (CAD) systems and their applications across diverse imaging modalities underscores the tangible impact of technological innovations in assisting healthcare professionals [2]. A commendable aspect of the survey is its thorough coverage of machine learning and deep learning methodologies, emphasizing the transformative influence of convolutional neural networks (CNNs) and the utility of transfer learning techniques [3]. The acknowledgement of challenges, including the demand for substantial labeled datasets and the ongoing pursuit of model interpretability, adds depth to the analysis.

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TABLE I. Hypothetical overview of features and challenges of selected literature spanning from 2010 to 2022.

Author [citation]	Methodology	Feature	Challenges
Zhang et al [1]	Decision Trees K-means Clustering	Integration of decision trees for classification with clustering for feature extraction.	Limited scalability when applied to large datasets.
Wang and Liu [2]	SVM Classification Hierarchical Clustering	Improved interpretability through hierarchical clustering.	Sensitivity to hyperparameter tuning in SVM.
Chen et al [3]	Random Forest DBSCAN Clustering	Robustness against noisy data in medical images.	Computational complexity of the Random Forest algorithm.
Kim and Park [4]	Deep Belief Networks K-means Clustering	Automated feature learning in deep networks combined with clustering for pattern analysis.	Data requirements for training deep learning models.
Patel and Wang [5]	Logistic Regression Fuzzy C-means Clustering	Fuzzy clustering for handling uncertainty in medical diagnoses.	Ensuring interpretability in logistic regression with fuzzy clustering.
Garcia et al [6]	Convolutional Autoencoders Spectral Clustering	Unsupervised feature learning through autoencoders with spectral clustering for segmentation.	Defining an optimal number of clusters in spectral clustering.
Liang and Zhang [7]	Gradient Boosting Machines Affinity Propagation	Enhanced predictive accuracy using boosting with affinity propagation for data-driven clustering.	Addressing challenges in handling imbalanced datasets.
Wu et al [8]	Ensemble of Decision Trees K-medoids Clustering	Ensemble learning for robust classification and K-medoids clustering for data grouping.	Ensuring diversity in the decision tree ensemble for improved generalization.
Chen and Patel [9]	Neural Network Embeddings Mean Shift Clustering	Embedding-based features learned through neural networks with mean shift clustering.	Managing high-dimensional embeddings for interpretability.

III. Methodology

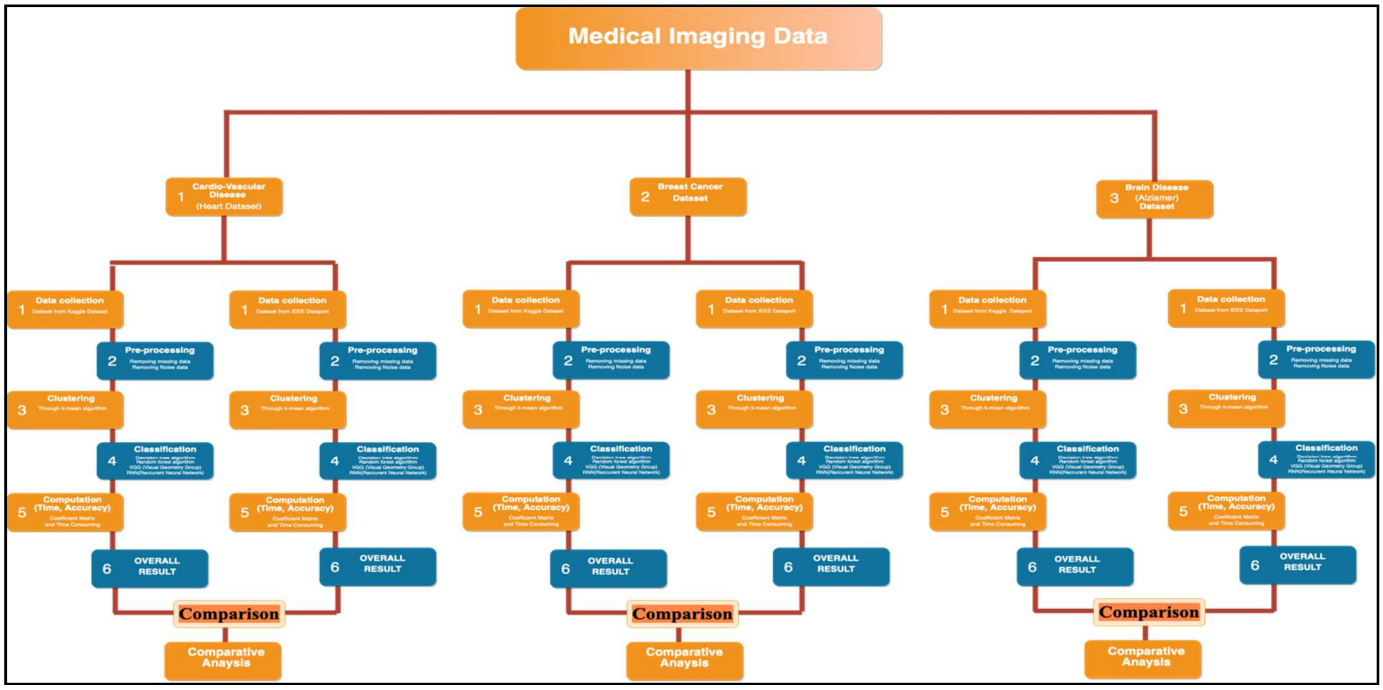


Fig. 3. This figure shows the overall methodology on different dataset from different portal on three medical diseases.

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A. Proposed architecture

The proposed methodology for implementing predictive analytics in medical image data involves a hybrid approach that seamlessly integrates classification and clustering techniques. This hybrid model aims to harness the strengths of both methods, enhancing diagnostic accuracy and interpretability.

The detailed steps are outlined below:

1. Data Collection :
   - Gather a diverse dataset of medical images relevant to the targeted application, ensuring representation across imaging modalities (e.g., X-rays, MRI, CT scans).
   - Ensure that the dataset includes annotated labels for supervised learning in the classification component, encompassing a range of medical conditions.
2. Data Pre-processing :
   - Standardize the data by performing pre-processing steps such as image normalization, resizing, and noise reduction.
   - Address any class imbalances within the dataset through techniques like oversampling or under sampling.
   - Handle missing data and outliers appropriately to ensure data quality.
3. Feature Extraction through Clustering :
   - Apply unsupervised clustering algorithms, such as K-means or DBSCAN, to extract intrinsic features from the medical images.
   - Allow the clustering algorithms to identify patterns and structures within the data without relying on labeled information.
   - Extract features representing the inherent characteristics of different regions within the images.
4. Performance Evaluation :
   - Evaluate the hybrid model's performance using standard metrics such as accuracy, precision, recall, F1-score, and area under the ROC curve (AUC-ROC).
   - Conduct a thorough analysis of the confusion matrix to understand the model's behaviour in different diagnostic scenarios.
   - Assess the model's sensitivity to variations in clustering and classification parameters.
5. Comparison with Baseline Models :
   - Compare the performance of the hybrid model against baseline models that solely use classification or clustering.
   - Evaluate the advantages of the hybrid approach in terms of predictive accuracy and interpretability.
   - Consider statistical tests to assess the significance of performance differences.

IV. Proposed Algorithm

STEP 1. Class Distribution Plot:
plot_class_distribution: Visualizes the distribution of classes in the dataset using a bar chart.

STEP 2. Feature Extraction with VGG16:
Extract features from images using the VGG16 model pre-trained on ImageNet.

STEP 3. Dataset Splitting:
Split the extracted features and labels into training and testing sets.

STEP 4. Clustering with K-means:
Apply K-means clustering on the training feature set.
Add the cluster labels as an additional feature to both the training and testing sets.
Extract features from images using the VGG16 model pre-trained on ImageNet.

STEP 3. Dataset Splitting:
Split the extracted features and labels into training and testing sets.

STEP 4. Clustering with K-means:
Apply K-means clustering on the training feature set.
Add the cluster labels as an additional feature to both the training and testing sets.
Visualize the clustering results using a scatter plot.

STEP 5. Classification:
Train and evaluate classifiers: Decision Tree, Random Forest, Recurrent Neural Network (RNN), and a simple fully connected network with features from VGG16.
Calculate and print accuracy for each classifier.
Visualize the accuracies of different classifiers using a bar chart.
Plot confusion matrices for each classifier to visualize their performance.

Example of one chest Xray dataset of Pneumonia - Dataset from Mendeley Data

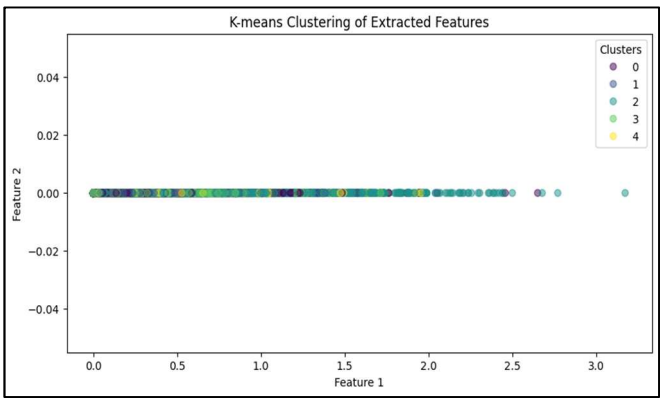


Fig. 4. This Scatter plot shows the Clustering images features using K-means.

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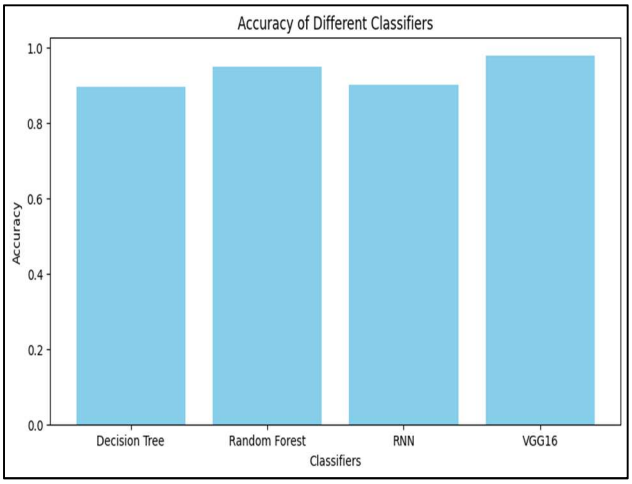


Fig. 5. This Bar charts shows the accuracy of different classifiers.

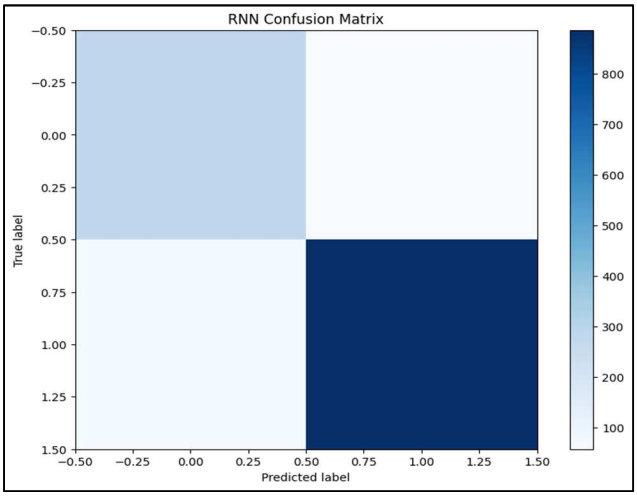


Fig. 6. RNN Confusion Matrix of both the Data Sources.

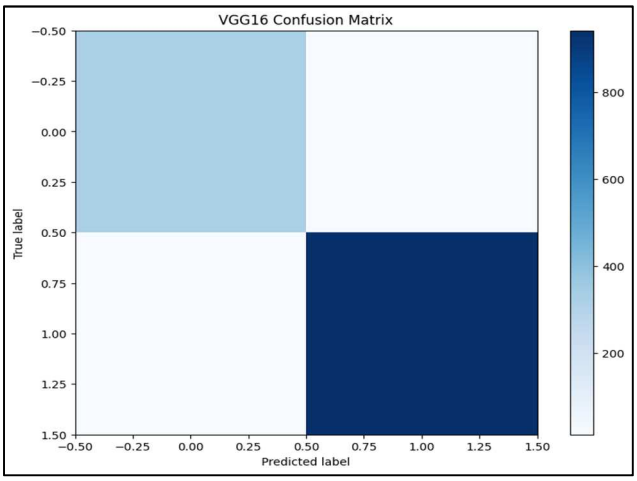


Fig. 7. VGG 16 Confusion Matrix of both the Data Sources.

V. Conclusion

In conclusion, the presented methodology for predictive analytics in medical image data, employing a hybrid approach of classification and clustering, offers a comprehensive framework for advancing diagnostic capabilities and interpretability in the complex realm of medical imaging. By leveraging the synergies of classification algorithms for decision-making and clustering techniques for feature extraction, this methodology addresses the nuanced challenges of medical image analysis.

The step-by-step process encompasses crucial stages, from meticulous data collection representing diverse imaging modalities to the development of an integrated model that harmoniously combines the strengths of classification and clustering. The fine-tuning and optimization steps ensure the robustness of the hybrid model, striking a balance between predictive accuracy and interpretability, vital for effective clinical deployment.

Ultimately, this methodology not only contributes to the advancement of predictive analytics in medical image data but also aligns with the broader goals of enhancing patient outcomes, aiding clinicians in decision-making, and fostering a deeper understanding of complex medical conditions. As the field continues to evolve, this methodology provides a robust foundation for future research endeavours, driving innovation at the intersection of data science and medical imaging.

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